GE VERNOVA

GridOS Simulation Framework

0.21.0

User Guide

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Change History

Date	Change Description
2024-Sep-17	GridOS Simulation Framework 0.1.0 (24R09): • This is a new document for the initial internal release of GridOS Simulation Framework.

2024-Oct-10	GridOS Simulation Framework 0.2.0 (24R10):
	• Organizer with Simulation Plugin is now Simulation Manager.
	• Ingress URL changed
	• Simulation Manager API
	  ○ Added error endpoint
	  ○ Updated deploy endpoint
	  ○ Removed loaddata, init, and prime endpoints

2024-Nov-05	GridOS Simulation Framework 0.3.0 (24R11): • Scenario Management ○ Validity included when listing scenarios
2024-Nov-25	GridOS Simulation Framework 0.4.0 (24R12): • Scenario state values renamed

2024-Dec-18	GridOS Simulation Framework 0.5.0 (24R13):
	• Simulation Manager API
	  ○ Status response include overall state for the instance and each Simulator/Application
	  ○ Optional description for simulation scenarios

2025-Jan-24	GridOS Simulation Framework 0.6.0 (25R01):
	• `simulation` namespace replaced with `foundation-env-default`
	• `simulation-worker` namespace replaced with `simulation-grid-operate`, `simulation-grid-plan` and `simulation-grid-transact`
	• Helm umbrella charts renamed:
	  ○ `simulation-charts` renamed to `simulation-apps`
	  ○ `simulation-worker-charts` renamed to `simulation-support`

Date	Change Description
2025-Feb-27	GridOS Simulation Framework 0.7.0 (25R02): • Allow simulator only simulation scenarios • Check `contentType` for archives when validating simulation scenarios • Store simulation events to Timebase
2025-Mar-24	GridOS Simulation Framework 0.8.0 (25R03): • Remove support for UAA
2025-Apr-28	GridOS Simulation Framework 0.9.0 (25R04): • Simulation Manager REST API path changed to `/simulation/manager/v1` • `simulationTimestamp` on deploy is used as the reference timestamp. • `/simulation/manager/v1/instances` API is added to get all simulation instances. • Add Authorization to Simulation Manager API

2025-May-08	GridOS Simulation Framework 0.10.0 (25R05):
	• Source updated in example simulation events.
	• Simulation Manager REST API
	  ○ Revised response for instances
	  ○ Simulation instance and scenario names are limited to 52 characters.
	  ○ Simulation scenario descriptions are limited to 250 characters.

2025-May-30	GridOS Simulation Framework 0.11.0 (25R06): • Simulation Manager REST API ○ Return an error if a user attempts to delete a scenario that is in use by a simulation instance.
2025-Jun-30	GridOS Simulation Framework 0.12.0 (25R07): • Simulation Manager handling of `ERROR` during simulation instance deployment.
2025-Aug-04	GridOS Simulation Framework 0.13.0 (25R08): • User Guide updated to remove detailed REST API request/response examples and error handling. • All API call details moved to the Programmer Guide under Simulation Manager REST API documentation. • User Guide now provides high-level summaries and cross-references to Programmer Guide for API • Update for implementation of `FIXED_SINGLETON` locks after simulation instance state migration to PostgreSQL
2025-Sep-03	GridOS Simulation Framework 0.14.0 (25R09): • Move simulation events and states to Programmer Guide. • Move deployment type details to Programmer Guide. • Add pause on error.
2025-Oct-10	GridOS Simulation Framework 0.15.0 (25R10): • Include warning note about archive manager exclusion in scenario list page validity.

Date	Change Description
2025-Nov-05	GridOS Simulation Framework 0.16.0 (25R11): • Archive metadata now supports user property in addition to content type. • Quick filters for simulation instances. • Event logs for simulation instances.
2025-Dec-02	GridOS Simulation Framework 0.17.0 (25R12): • Real time updates for simulation instance event logs. • Quick filters for simulation scenarios.

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1. GridOS Simulation Framework Introduction

To match our composable and modular GridOS applications, our GridOS simulations must also be composable and modular. Each customer deployment could be unique in terms of which GridOS applications are deployed, and the GridOS simulation environment has to match that.

To meet these needs we require a GridOS Simulation Framework that supports:

- Integrating and coordinating across multiple GridOS simulators and applications
  - Multiple products in a single simulation environment, such as Markets + EMS, EMS + DMS, EMS + WAMS.
- A single control plane for simulation that works across the entire solution, regardless of traditional solution boundaries.
- Easy definition of a simulation scenario
  - Selection of GridOS simulators and applications
  - Model and data set selection
- Easily run multiple simultaneous instances of a simulation scenario
  - Allowing scale out in cloud environments
- Interfacing with applications that could be running outside of GridOS
- Interfacing with 3rd party simulators and applications
- API driven in addition to being UI driven
- Integration with AI/ML ops so GridOS simulation can be used to test new AI/ML models

1.1. Simulation Manager UI

The Simulation Manager UI is available at the URL https://service.hostname/simulation/ui/ where GridOS Simulation Framework is installed. To access the UI, you must have one of the read permissions listed at authorization.

1.2. Running simulations in GridOS

A simulation scenario is a particular combination of simulators and applications to run together with a particular model and data set(s).

Scenario management supports creating, editing and deleting the simulation scenarios. GridOS simulators and applications have to follow some integration requirements to be available for use in a simulation scenario.

A simulation instance is then deployed on a GridOS simulation environment before being executed.

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2. Authorization

GridOS Simulation Framework defines a number of permissions that are required for calling the Simulation Manager REST API or accessing the Simulation Manager UI.

The following permissions in `request.auth.claims[sws_permissions]` claim of the JWT token are supported.

Table 1. GridOS Simulation Framework Permissions

Permission Name	Description
sim.scenario.read	Read access to my scenarios
sim.scenario.readAll	Read access to all scenarios
sim.scenario.write	Write access to my scenarios
sim.scenario.writeAll	Write access to all scenarios
sim.instance.read	Read access to my instances
sim.instance.readAll	Read access to all instances
sim.instance.execute	Execute access to my instances
sim.instance.executeAll	Execute access to all instances

In the descriptions above, `my` refers to the user who owns the scenario or instance. The `all` permissions are for all scenarios or instances created by any user.

GridOS Foundation roles and permissions **must** be configured to ensure that user groups include the required GridOS Simulation Framework permissions. These user groups can then be associated with UI or API users.

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3. Scenario Management

Simulation scenarios define the set of simulators and applications that can be deployed on GridOS Simulation Framework. You can create, edit, duplicate, and delete scenarios using the GridOS Simulation UI.

3.1. Managing Scenarios via the UI

1. Navigate to the **Scenario List** tab to view all scenarios.
2. To create a new scenario, click the **Create Scenario** button and fill in the required fields in the dialog box.
3. To edit an existing scenario, click the **Edit** icon in the Actions column or open the scenario and click **Edit** in the Scenario Details tab.
4. To duplicate a scenario, click the **Copy** icon in the Actions column and complete the Duplicate Scenario dialog.
5. To delete a scenario, click the **Delete** icon in the Actions column and confirm in the popup. If the scenario is in use, an error message appears.
6. Click on a scenario row to view details in the **Scenario Details** tab, including status, simulators, applications, and archives.
7. To deploy a scenario, click **Deploy Simulation** in the Scenario Details tab and complete the Deploy Simulation dialog.

The scenarios table can be filtered easily by username using the control above the table.

For more details on the available controls and dialogs, see:

- Scenario List tab
- Scenario Details tab

3.2. Scenario Validity

For a simulation scenario to be successfully deployed, it must be valid. The following conditions must be true:

- At least one simulator is included
- All application and simulator custom resources are available on the GridOS cluster
- The scenario state is `Published`
- Any archives listed in the scenario
  - Must exist in Archive Manager
  - Must match the archive metadata specified in the custom resource. Either the `contentType`

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or the `userProperty` must match.

The validity of a scenario is shown in the Scenario List and Scenario Details tabs.

> The scenario list page validity does not check the Archive Manager to check the validity of the archives inside the simulator and applications.

3.3. Scenario Operations

All scenario management actions (create, edit, duplicate, delete, deploy) can be performed using the UI as described above.

For technical details on managing scenarios using the REST API, see Simulation Manager REST API documentation.

4. Deployment

You can deploy simulation scenarios in a GridOS Simulation environment. Multiple instances of the same scenario can be deployed by different users.

To deploy a simulation instance using the GridOS Simulation UI:

1. Navigate to the **Scenario Details** tab.
2. Click the **Deploy Simulation** button.
3. In the Deploy Simulation dialog box, enter a name for your simulation instance and select a reference time (Realtime or Custom).
4. Click **Initialize** to deploy the simulation.

The simulation instance is deployed and initially shows as `Deploying`. Simulation Manager uses simulation events to transition the instance through the following stages:

- `Deploying`
- `Deployed`
- `Data Loaded`
- `Initialized`
- `Ready to Start`

If at any point during deployment a simulator or application reports `Error`, the deployment process pauses. You must resolve the error in the appropriate simulator or application so they report the correct state. This allows the deployment to continue.

The user who deploys an instance becomes the owner of that instance.

If the scenario is invalid, deployment does not proceed and you see an error message in the UI.

`FIXED_SINGLETON` simulators or applications can only be associated with a single simulation instance ID at any time. If you attempt a new deployment where a `FIXED_SINGLETON` is already in use, the UI displays an error and deployment fails.

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5. Execution

5.1. Managing Instances via the UI

1. Navigate to the **Instances List** tab to view all instances.
2. To delete an instance, click the **Delete** icon in the Actions column and confirm in the popup.
3. Click on a instance row to view details in the **Instance Details** tab.

The instances table can be filtered easily by username or status (`Ready`, `Running` or `Error`) using the controls above the table.

For more details on the available controls and dialogs, see Simulation List tab.

5.2. Instance Details

After deployment, the simulation instance is in the `Ready` state.

To execute, pause, or inspect a simulation instance using the GridOS Simulation UI:

1. Navigate to the **Simulation Details** tab for your simulation instance.
2. To start the simulation, click the **Start Simulation** button when the status is `Ready`.
3. To pause a running simulation, click the **Pause Simulation** button.
4. To end the simulation, click the **End Simulation** button (only visible for users with execute permissions).

The **Overview** tab shows key information about the simulation instance:

1. The status pill at the top of the page shows the current state of the simulation (e.g., `Ready`, `Running`, `Paused`, `Error`).
2. You can view detailed status and sub-service information in the tables and panels on this tab.

The overall state reported for either the simulation instance or a simulator/application is based on the order of the simulation state transition. The earliest state across all services is reported as the overall state. If any service reports `Error`, the overall state is `Error`.

The **Logs** tab shows the recent simulation events associated with this instance.

1. By default the last hour of events are shown. You can select from a fixed set of time ranges between the last 15 mins and 24 hours.
2. The event log is updated in real time as new events occur.
3. A quick filter allows only error events to be shown.

> Due to limitations in the Timebase REST API, the event logs can only show a maximum of 1000 events regardless of how many events are stored for the time period selected.

For more details on the available controls and status indicators, see Simulation Details tab.

For technical details on executing, pausing, listing, and retrieving simulation instances using the REST API, see Simulation Manager REST API documentation.

5.3. Pause On Error

Individual components of a Simulator or Application can be configured to pause the entire simulation instance automatically if they enter the `Error` state. This is controlled by the `pauseOnError` flag in the custom resource.

When `pauseOnError` is set to `true` for a component, if that component transitions to the `Error` state during execution, the entire simulation instance automatically transitions to the `Paused` state. You can then investigate and resolve the error in the Simulator or Application so that it reports the correct `Paused` state. Once all components are in the `Paused` state, you can resume the simulation instance.

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6. Deletion

To delete a simulation instance using the GridOS Simulation UI:

1. Select the **Simulation List** tab to view all simulation instances.
2. Locate the instance you want to delete.
3. Click the **Delete** icon (trash can) in the Actions column for that instance.
4. Confirm the deletion in the popup dialog.

You can also delete an instance from the **Simulation Details** tab by clicking the **End Simulation** button at the top of the page.

Deleting an instance removes its state, deletes any `ON_DEMAND` resources, and frees any `FIXED_SINGLETON` resources used. The deletion process publishes the destroy event to all GridOS services involved in the simulation instance.

> Only users with instance execute permissions can see the delete options.

For technical details on deleting a simulation instance using the REST API, see the Simulation Manager REST API documentation.

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